Calorimetry

1. A copper sample (c_{Cu} = 387 J/(kg K)) of mass m_c = 75g and temperature T_c = 312 °C is dropped into a glass beaker that contains a mass of water m_w = 220g (c_2 = 4190 J/(kg K)).

   The *heat capacity* (the specific heat times the mass) of the beaker is C_b' = 190 J/K. The initial temperature of the water beaker is $T_{w,b}$ = 21.0 °C.

   What is the final temperature of the copper, beaker, and water?

   Solution:

$$\Delta U_c + \Delta U_w + \Delta U_b = 0 \tag{1}$$
$$mc_{Cu}(T - Tc) + m_w c_w(T - T_w) + C_b(T - T_b) = 0 \tag{2}$$
$$T = 301\text{K} \tag{3}$$

2. (a) A metal rod of mass 1 kg, specific heat 1000 J/kg°C, initial length 10 cm, and initial temperature 20 °C is placed on a hot plate. Heat flows into the rod with a power of 200 W. After a time t, the metal rod has reached thermal equilibrium of 120 °C and now has a length of 10.01 cm. Determine the time t and the coefficient of linear expansion α.

   Solution: First, we need to remember from 7A that power P (don't confuse this with pressure!) is a change in energy over time, ie $P = \frac{\Delta E}{\Delta t}$. We also know that heat flow Q is related to a change in temperature ΔT by $Q = mc\Delta T$.

$$Q = P\Delta t = mc\Delta T \tag{4}$$

   We can solve this for time t.

$$t = \frac{mc\Delta T}{P} = \frac{(1\,\text{kg})\,(1000\,\text{J/kg°C})\,(100\,°\text{C})}{200\,\text{W}} = 500\,\text{s} \tag{5}$$

   Now, to calculate the coefficient of linear expansion, recall that $\Delta L = L_0\alpha\Delta T$.

$$\alpha = \frac{\Delta L}{L_0\Delta T} = \frac{0.01\,\text{cm}}{(10\,\text{cm})\,(100\,°\text{C})} = 10^{-5}\,°\text{C}^{-1} \tag{6}$$

   (b) The rod is cooled to 100 °C and is then immersed in a mixture of ice and water initially in equilibrium. Initially, the ice has mass m and the water has mass M. Find a condition on m that will ensure the final equilibrium temperature T_f of the system is unchanged from the initial temperature of the ice+water mixture. (For simplicity, you may approximate the specific heat of water as 5000 J/kg∘C and its heat of fusion as 300000 J/kg).

   Solution: For the final equilibrium temperature to be equal to the initial temperature, all of the heat of the rod must go into melting the ice:

$$Q = |(m_r C_r(T_f - T_0)| < mL \tag{7}$$

   It is a mixture of ice and water in equilibrium. Therefore $T_0 = 0\,°\text{C}$.

$$m > \frac{(1\,\text{kg})\,(1000\,\text{J/kg°C})\,(100\,°\text{C})}{300\,000\,\text{J/kg}} = 0.33\,\text{kg} \tag{8}$$

   (c) Let $m = 0.1$ kg and $M = 0.4$ kg. Determine the final equilibrium temperature T_f of the system.

   Solution: $m < 0.33$ kg, so all the ice has melted. Therefore we have:

$$m_r c_r(T_f - T_{r0}) + (M + m)c_w(T_f - T_{w0}) + mL = 0 \tag{9}$$

   so that

$$T_f = \frac{(m + M)c_w T_{w0} + m_r c_r T_{r0} - mL}{m_r c_r + (m + M)c_w} = 20\,°\text{C} \tag{10}$$

First Law

3. A rubber ball of mass m, radius r, and specific heat c is rolled off a cliff 10 m above the ground. It bounces back up to a height of 9 m.

   (a) Assume all the energy lost in the collision goes into the ball. Has heat been added to the ball? Has work been done on it? What is the temperature of the ball after the collision?

   Solution: The gravitational potential energy that is lost during the bounce will be transmitted to the ball in the form of heat. Therefore,

   $$mc\Delta T = mg\left(h_f - h_i\right) \rightarrow \Delta T = \frac{g(h_f - h_i)}{c} \tag{11}$$

   (b) If the ball always bounces to 90% of its starting height, what is the temperature of the ball after it stops bouncing?

   Solution: Once the ball stops bouncing, all of its initial gravitational potential energy will be converted into heat.

   $$mc\Delta T = mg\left(h - r\right) \rightarrow \Delta T = \frac{g(h - r)}{c} \tag{12}$$

   (c) Suppose that we now take into account the expansion of the ball due to heating. If the linear expansion coefficient of the ball is α, the what is the final temperature?

   Solution: The thermal expansion of the ball will have the effect of raising its center of mass higher off the ground once it stops bouncing. Therefore the total potential energy being converted to heat will decrease slightly.

   $$mc\Delta T = mg\left(h - r - r\alpha\Delta T\right) \rightarrow \Delta T = \frac{g(h - r)}{c + gr\alpha} \tag{13}$$

   We can check our answer to part (c) by ensuring that we recover the answer to part (b) when we substitute in $\alpha = 0$, and, indeed, we do.

4. A washer (annulus) of mass m_w with inner radius a_0 and outer radius b_0 is loosely pinned to a surface by a screw of mass m_s with screw head of radius $r_0 > a_0$. How much heat should we apply to the washer screw combination if we wish to remove the washer without unscrewing the screw? Assume the washer and screw have specific heat capacities c_w and c_s, respectively, with $\alpha_w > \alpha_s$. Assume the change in temperature is small enough to warrant our use of linear thermal expansion.

   Solution: In order to remove the washer, we need to heat the system until the inner radius of the washer a is the same as the radius of the screw head r.

   $$a_0\left(1 + \alpha_w\Delta T\right) = r_0\left(1 + \alpha_s\Delta T\right) \tag{14}$$

   $$\Delta T = \frac{r_0 - a_0}{a_0\alpha_w - r_0\alpha_s} \tag{15}$$

   The amount of heat needed to cause the washer and screw to change temperature this much is

   $$Q = m_w c_w\Delta T + m_s c_s\Delta T \tag{16}$$

   $$= \left(m_w c_w + m_s c_s\right)\left(\frac{r_0 - a_0}{a_0\alpha_w - r_0\alpha_s}\right) \tag{17}$$

5. During a power failure you put a block of ice of mass $m = 20\,\mathrm{kg}$ at $0\,^\circ\mathrm{C}$ in the refrigerator, bringing the interior temperature to $0\,^\circ\mathrm{C}$. The rate of heat transfer per degree difference between the inside and the outside of the refrigerator is $R = 5\,\mathrm{W/K}$. Use $L_f = 300\,\mathrm{kJ/kg}$ as the latent heat of fusion of ice.

(a) Write the relationship between R, the temperature difference ΔT, the amount of heat Q exchanged and the corresponding time interval Δt, assuming Δt is small enough so that the ice does not melt.

Solution: $R\Delta t = \frac{Q}{\Delta T}$

(b) What is the amount of heat required for the ice to melt entirely?

Solution: $Q = mL_f = 6 \times 10^6\,\mathrm{J}$

(c) If room temperature is $20\,°\mathrm{C}$, how long will the ice last?

Solution: $\Delta t = \frac{Q}{R\Delta T} = 6 \times 10^4\,\mathrm{s} = 1000\,\mathrm{min} = 16$ hours 40 min

(d) What is the minimum amount of ice needed to have initially in order to remain at $0\,°\mathrm{C}$ for 20h ?

Solution: $mL_f = Q = R\Delta t\Delta T$, so $m = \frac{R\Delta t\Delta T}{L_f} = 24\,\mathrm{kg}$

6. A $2.0\,\mathrm{g}$ lead bullet moving at a speed of $200\,\mathrm{m/s}$ becomes embedded in a $2.0\,\mathrm{kg}$ block of a ballistic pendulum. Calculate the rise in in temperature of the bullet, assuming all the heat generated raises the bullet's temperature. The specific heat of lead is $0.16\,\mathrm{J/(g \cdot °C)}$.

Solution The initial kinetic energy of the bullet will be converted into the kinetic energy of the block-bullet system and the change in heat energy of the bullet. Thus,

$$\frac{1}{2}mv^2 = \frac{1}{2}(m+M)V^2 + mc\Delta T, \tag{18}$$

where M is the mass of the block, m is the mass of the bullet, V is the mass of the bullet-block system, and c is the specific heat of the bullet. We can determine V through conservation of momentum and then eliminate it from our energy conversion equation:

$$mv = (m+M)V \rightarrow \frac{1}{2}(m+M)V^2 = \frac{1}{2}\frac{m^2v^2}{m+M} \tag{19}$$

Solving for the change in the bullet's temperature, we obtain

$$\Delta T = \frac{1}{2c}\left(1 - \frac{m}{m+M}\right)v^2 \approx 125°C \tag{20}$$

Heat Transfer

7. A copper rod and an aluminum rod of length $L_{cu} = 20\,\mathrm{cm}$ and$L_{al} = 30\,\mathrm{cm}$ and same cross-sectional area A are attached end to end (see figure below). The copper end is placed in a furnace maintained at a constant temperature of $225\,°\mathrm{C}$. The aluminum end is placed in an ice bath held at constant temperature of $0.0\,°\mathrm{C}$. Assume that the thermal conductivity of aluminum and copper are given by $k_{al} = 5.0 \times 10^{-2}\,\mathrm{J/(s \cdot m \cdot °C)}$ and $k_{cu} = 10 \times 10^{-2}\,\mathrm{J/(s \cdot m \cdot °C)}$, respectively.

(a) Calculate the temperature at the point where the two rods are joined.

Solution: There is conservation of heat flow at the interface between the two bars.

$$H_{cu} = H_{al} \tag{21}$$

$$\frac{dQ}{dt}|_{cu} = \frac{dQ}{dt}|_{al} \tag{22}$$

$$k_{cu}A\frac{T_H - T_M}{L_{cu}} = k_{al}A\frac{T_M - T_L}{L_{al}} \tag{23}$$

$$T_M = \frac{\frac{k_{cu}}{L_{cu}}T_H + \frac{k_{al}}{L_{al}}T_L}{\frac{k_{cu}}{L_{cu}} + \frac{k_{al}}{L_{al}}} = 169\,^\circ\mathrm{C} \tag{24}$$

(b) What is the effective thermal conductivity k_{eff} of the system composed of these two rods ?

Solution:

$$k_{cu}A\frac{T_H - T_M}{L_{cu}} = k_{al}A\frac{T_M - T_L}{L_{al}} = k_{eff}A\frac{T_H - T_L}{L_{cu} + L_{al}} \tag{25}$$

$$k_{eff} = k_{cu}\frac{Lcu + Lal}{L_{cu}}\frac{T_H - T_M}{T_H - T_L} \tag{26}$$

$$k_{eff} = \frac{L_{cu} + L_{al}}{\frac{L_{cu}}{k_{cu}} + \frac{L_{al}}{k_{al}}} = 6.25 \times 10^{-2}\,\mathrm{J/(s \cdot m \cdot {}^\circ C)} \tag{27}$$

or, with $L_{eff} = L_{cu} + L_{al}$

$$\frac{L_{eff}}{k_{eff}} = \frac{L_{cu}}{k_{cu}} + \frac{L_{al}}{k_{al}} \tag{28}$$